

1 Part 1

Formulas

$$\sum_{i=0}^{n-1} ar^i = \frac{a(1-r^n)}{1-r}$$

For given rate $r\%$ over time period $t \in [0, T]$, with annual period P , discount factor $d_{0,T}$ is given

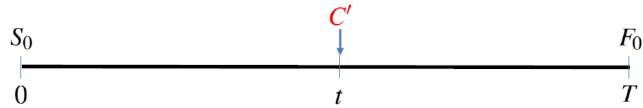
$$d_{0,T} = \left(1 + r\% \times \frac{T}{P}\right)^{-1} < 1$$

No Arbitrage Argument

To prove $A = B$, show strategies for $A \neq B$. If $A > B$, short on A , long on B , and vice versa.

1.1.2 Asset w/ Predictable Income

$$F_0 = (S_0 - C)/d_{0,T}, \text{ where } C = C' d_{0,t}$$

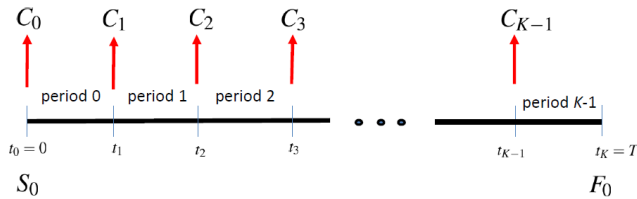


1.1.3 Asset w/ Continuous Dividend Yield

Annualized continuous dividend yield of q

$$F_0 = S_0 \exp((r - q)T)$$

1.1.4 Asset w/ Storage Costs

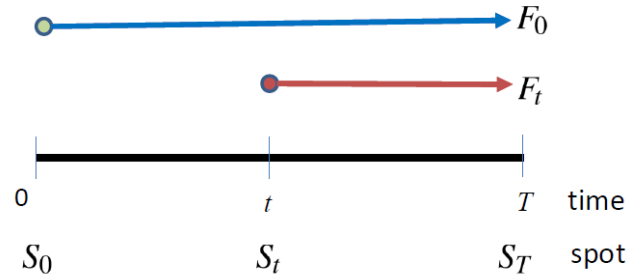


$$d_{0,K} F_0 = S_0 + \sum_{j=0}^{K-1} d_{0,j} C_j$$

1.1.5 Convenience Yield

$$d_{0,K} F_0 = S_0 + \sum_{j=0}^{K-1} d_{0,j} (C_j - y)$$

1.2 Value of Forward



$$f_t = (F_t - F_0) d_{t,T}$$